

RITIK KUMAR

4th Year, BS+MS (Research) Physics
Indian Institute of Science, Bengaluru

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EDUCATION

Indian Institute of Science, Bengaluru, India.
(BS+MS) in Physics

Aug. 2022 – July 2027 (expected)

RESEARCH EXPERIENCE

I work in high-energy astrophysics with a focus on active galactic nuclei (AGN), using broadband X-ray spectroscopy to study coronal physics, accretion variability, and outflow processes. My research combines multi-epoch NuSTAR, XMM-Newton, optical (SDSS) observations with physical spectral modelling in XSPEC, including reflection models (pexrav, xillver/xillverCp, relxill), Comptonisation models (nthcomp), and ionised absorber/emitter models (zxcipcf, warmabs, pion with velocity shifts). I use these tools to measure high-energy cutoffs, coronal temperatures, reflection fractions, and ionised absorption features linked to winds and feedback. I also work on machine learning models for light curve reconstruction of gamma-ray bursts. I am particularly interested in the spectral evolution of Changing-Look AGN, especially how variations in accretion rate, obscuration, and disk-corona geometry manifest in the hard X-ray band. I have developed some knowledge on timing analysis and eager to explore more on timing analysis. My broader goal is to understand the physical drivers of AGN state transitions by integrating broadband X-ray diagnostics with multi-wavelength signatures and timing analysis.

Tracing Outflows Across Different Gas Phases in AGN

Sep. 2025 – Apr. 2026

Bachelor's Thesis

Advisor: **Prof. C. S. Stalin**, IIA, Bengaluru

- Systematically investigating ionized outflows in AGN through X-ray analysis of sources from the Milli Quasar Catalog cross-correlated with XMM-Newton archives. I cross-correlated the milli-quasar catalog with XMM-Newton at 5".
- Combining X-ray detections to explore connections between different outflow phases and operational scales in AGN
- We identified several AGN for which detailed outflow studies are not yet available; currently performing spectral modelling of their XMM-Newton data and we discovered several systems with multiple outflow components at 0.06–0.33c—rare “UFO forests” challenging simple wind models and we are still conducting full analysis.
- Modeling methodology: Began with a simple power-law continuum, then incorporated reflection components to isolate residual features associated with outflows. After identifying line-like signatures, Gaussian absorption/emission models (gabs, zgauss) were applied for detection. Finally, physical modeling was performed using zxcipcf and XSTAR grids for the final spectral analysis.
- We have found the sources which are showing the warm absorption and UFO from Fe K line. And interesting thing is, we have found 2 to 3 source for the first time and these sources have soft x-ray ultrafast outflows like the paper by J N Reeves found in PDS 456. To find these signature and model them is not possible with available models like zxcipcf, therefore, I developed the XSTAR grid to model these signature.
- Manuscript is in preparation based on the source "****"** where we have found soft x-ray ultra-fast outflows and Fe K line outflows together using XSTAR (photoionised grid) and this is case of **multiphase outflows**.: soft x-ray UFO velocity 0.22c and Fe K line UFO velocity is 0.32c.

Spectral Analysis of Seyfert Galaxies

Dec. 2024 – Sep. 2025

Advisor: **Prof. C. S. Stalin**, IIA, Bengaluru

This is my independent work with Prof C. S. Stalin.

- Developed a Python pipeline to cross-match NLSy1/BLSy1 catalogs with NuSTAR observations (12 radius) using angular-separation and signal-to-noise criteria.
- Reduced NuSTAR data with NuSTARDAS (FPMA/FPMB), including calibration, background filtering, and barycentric corrections.
- Performed broadband spectral modeling in XSPEC using cutoffpl, nthcomp, pexrav, xillverCp, and relxill to constrain coronal temperature, high-energy cutoff, and reflection properties.

- Measured coronal temperatures across multiple epochs with well-constrained uncertainties, finding evidence for temporal stability in several sources.
- **Reported coronal temperature measurements for multiple sources for the first time. And setup the geometry, thickness of corona, reflection fraction, variability and one of the important source is ***** where we have seen the coronal temperature is varying over epochs means disk is not stable.**
- *Manuscript in preparation.*

Lead Researcher – Relativistic Reflection Modeling in AGN Jan. 2026 – Present (Ongoing)

Full Member, ECAP X-ray Research Group

Collaboration with **Prof. Thomas Dauser, Prof. Jörn Wilms, Dr. Alexey Nekrasov**

Erlangen Centre for Astroparticle Physics (ECAP), Friedrich-Alexander-Universität, Germany

- Leading an independent research project on developing and testing new relativistic reflection models for AGN X-ray spectroscopy, initiated through direct outreach to Prof. Thomas Dauser
- **Granted full membership in the ECAP X-ray research group (March 2026)**, including:
 - * Full SSH access to Remeis Observatory computing cluster for large-scale data analysis
 - * Active membership in Sternwarte X-ray research mailing list
 - * Regular participation in weekly ECAP X-ray group research meetings
- Serving as lead researcher: Got the source, performed complete data reduction pipelines (calibration, filtering, spectral extraction) for XMM-Newton and NuSTAR, and conducting broadband spectral modeling under guidance from ECAP group.
- Testing and validating new relativistic reflection model variants (extended corona, ring-like corona given by [Nekrasov, A. D, et al. 2025](#)), with real observational data, analyzing reflection signatures, disk-corona geometry, height of corona, and spin measurements through detailed XSPEC modeling
- Demonstrates initiative in building international research collaborations, capability to conduct independent publication-track research, and integration into world-leading X-ray astrophysics research group

Light Curve Reconstruction of Gamma-Ray Bursts

Apr. 2025 – Present (Ongoing)

Advisor: Prof. Maria G. Dainotti, NAOJ, Japan

Collaborative group project. My contributions:

- Developed and optimized Deep Gaussian Process pipelines for reconstruction of gamma-ray burst (GRB) light curves.
- Applied Deep Gaussian Process models to 545 GRBs, including highly sparse and noisy datasets with limited observational coverage.
- Developed methods for uncertainty estimation and robust inference in irregular astrophysical time-series data while minimizing false detections.
- Contributed to computational workflow optimization, model comparison, and statistical analysis across multiple machine learning reconstruction frameworks.
- Collaborated with an international research group and participated in regular scientific discussions on GRB time-domain analysis.
- Kaushal, Aditi, M.G. Dainotti, **Ritik Kumar, et al.** “Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves,” *Journal of High Energy Astrophysics (JHEAp)*
DOI: [10.1016/j.jheap.2025.100519](https://doi.org/10.1016/j.jheap.2025.100519)
- Currently, we are working on our next paper of this series using different ML models.
- Research conducted under the mentorship of Prof. Maria G. Dainotti in an international collaboration; **testimonial available upon request.**

Physics-Informed Neural Networks for ODEs/PDEs

Jan. 2025 – Apr. 2025

Advisor: Prof. Rishita Das, IISc, Bengaluru

- Implemented PINNs in PyTorch to capture discontinuities in shock tubes.
- Compared PINNs with traditional numerical methods (Runge-Kutta, Euler Method, central difference), which proved more reliable for handling discontinuities.
- Solved 1D/2D/3D Burgers’ equations using different initial and boundary conditions, Poisson equation. We matched our results of PINNs with Numerical method and they were nearly the same.
- Solved the 1D Euler’s equation using PINNs with initial Sod Shock Tube, and we captured partial shocks.
- Applied CFL stability criterion ($CFL = 0.5$) with 400 grid points for stable time-stepping in compressible flow simulations with shock wave propagation.

PUBLICATIONS & PRESENTATIONS

Journal Articles

- Kaushal, Aditi, M. G. Dainotti, **Ritik Kumar**, et al. (2025). *Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves*. **Journal of High Energy Astrophysics (JHEAp)**. DOI: [10.1016/j.jheap.2025.100519](https://doi.org/10.1016/j.jheap.2025.100519)

Manuscripts in Preparation

- **Ritik Kumar**, C. S. Stalin, and J. Hota, "Variation in the Temperature of the Corona in Seyfert-type AGN And disk geometry," 2026.
- **Ritik Kumar**, C. S. Stalin, and J. N. Reeves, "A possible multi-phase outflows in SMSS J114447.77-430859.3."

Conference Presentations

- **XRISM International Conference 2025, Japan** — [Poster Presentation](#)
A comparative analysis of the coronal properties of Broad Line and Narrow Line Seyfert 1 galaxies.
- **Astronomical Society of India, Conference 2026, Guwahati** — [Poster Presentation](#)
Nature of Coronae in Bright Active Galactic Nuclei and affecting the surrounding medium.

RELEVANT COURSES

Joint Astronomy Programme (JAP) – PhD-level collaborative program (IISc, ISRO, IIA, RRI)

- * PH 217: Fundamentals of Astrophysics
- * PH 362: Radiative Process in Astrophysics
- * PH 363: Introduction to Fluid Mechanics and Plasma Physics (Audit)
- * PH 354: Computational physics
- * PH 365: Galaxies and Interstellar Medium
- * PH 371: General Relativity and Cosmology
- * PH 377: Astronomical Techniques (Seminar Course)

WORKSHOPS

Fast Radio Bursts (FRBs) Workshop Oct. 6–17, 2025

International Centre for Theoretical Sciences (ICTS), India

Advanced workshop on FTSky: transient signal detection, radio data analysis, and multi-wavelength time-domain astrophysics.

Galaxies and Related Topics: Honoring Prof. Biman Nath Jun. 6–7, 2025
RRI, India

TECHNICAL SKILLS

X-ray Analysis: XSPEC, ISIS, HEASoft, NuSTARDAS, SAS; broadband spectral modelling and timing analysis.

Programming: Python (NumPy, SciPy, Astropy, Matplotlib, Pandas).

Data Pipelines: Python-based automation for NuSTAR/XMM reduction, light curve extraction, spectral extraction, multi-epoch AGN analysis.

Statistical/Machine Learning: Deep Gaussian Processes, Physics-Informed Neural Networks (PINNs), ReFANN, Numerical methods, uncertainty modelling. Time-series inference for irregular and noisy data; model comparison under stochastic variability, MCMC.

Additional Tools: dsplot, DS9, fv, topcat, Astroquery, SDSS (optical) data access, LaTeX.

LEADERSHIP AND SERVICE

- * **Management Coordinator**, Pravega (India's largest science and cultural festival) 2024 – 2025
Indian Institute of Science, Bengaluru.
- * **Member**, IISc Astrophysics Club "Astrae" arXiv Club of IISc.
- * **Member**, AstroInformatics Association
- * **Active participant**, arXiv Club journal discussions

ACHIEVEMENTS AND INTERNSHIPS

- * JEE Advanced 2022 – All India Rank 3424 (Top 0.3 % of 1,60,000 candidates)
- * 2025 Summer internship at **National Astronomical Observatory of Japan (NAOJ)** with **Prof. Maria Giovanna Dainotti**.
- * IIA: Since December 2024.